A CHARACTERISTIC-TWO $\mathbb{Q}$-FACTORIAL THREE-DIMENSIONAL PLT PAIR WITH A NON-NORMAL DIVISORIAL CENTRE

ARGUS OPEN RESEARCH PROJECT

Abstract. We document the counterexample of Cascini and Tanaka showing that in dimension three and characteristic $p = 2$, a $\mathbb{Q}$-factorial purely log terminal (plt) pair (Z, E) with E a prime divisor can have E non-normal.

1. Main Refutation

Theorem 1.1 (Cascini–Tanaka)**.** *Over an algebraically closed field of characteristic $p = 2$, there exists a projective $\mathbb{Q}$-factorial three-dimensional variety Z and a prime divisor $E \subset Z$ such that the pair (Z, E) is purely log terminal (plt), yet E is not normal.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 20000256 in the domain of birational geometry in positive characteristic follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *published refutation* deciding the scope: normality of prime plt centres in dimension 3 and characteristic 2 (Cascini-Tanaka).

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- characteristics p > 2.

References

[1] Primary problem list and source literature for Result 20000256 (birational geometry in positive characteristic), archived with exact theorem locators and verification certificates in the accompanying repository package.

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